

AI in Risk Management – Seminar Cheat Sheet

Definition

AI in risk management uses data and algorithms to identify, assess, monitor, and respond to risks more efficiently.

Core Process (memorise)

- Identify risks
- Assess risks (likelihood + impact)
- Monitor risks (real-time)
- Respond (alerts/automation)

Benefits

- Fast decision-making
- Handles big data
- Predictive insights
- Real-time monitoring
- Reduced human error

Key Risks (B-BAD)

- Bias (unfair outcomes)
- Black box (no transparency)
- Accountability issues
- Data quality problems

Mitigation Strategies

- Explainable AI (XAI)
- Human-in-the-loop
- Diverse datasets
- Bias testing & auditing
- Participatory design
- Governance & accountability

Case Studies

Greece (Kalogiannidis et al., 2024): AI for decision-making and risk analysis in organisations.

KAIF + IBM QRadar: AI-driven cybersecurity risk detection and monitoring.

Ready Sentences

- AI improves efficiency but introduces risks to fairness and accountability.
- Algorithmic bias can reinforce existing inequalities.
- Transparency and human oversight are essential.
- AI should support decision-making, not replace it.

Key References

- Hancock et al. (2024)
- Burrell (2016)
- O'Neil (2016)
- Floridi et al. (2018)
- Hubbard (2020)